

The image features the text "DL for ENG" in a bold, 3D, gold-colored font. The letters are floating on a deep blue ocean with visible ripples and small waves. In the background, there are rolling blue hills or mountains under a sky filled with soft, grey clouds. The overall scene has a serene, digital aesthetic.

DL for ENG

## LECTURE 8.1

# Stationary Heat

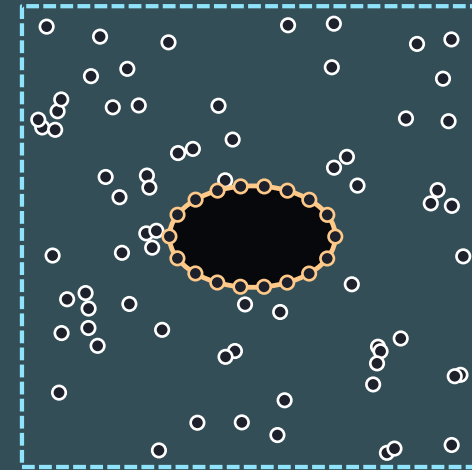
**Remus Teodorescu**

*With support from Research Assistant Noman Khan -  
nomank@energy.aau.dk*

Aalborg University · ret@et.aau.dk

Deep Learning for Engineering · 2026

## CONDUCTION · GEOMETRY · THE PLATE



*Steady conduction is Poisson's equation with units: Fourier's law, a conductivity, and walls held at a temperature, a flux or a convection. On a plate with a cooling hole the hole is built into the network by its level set, and the answer is checked by an energy balance.*

# Learning Objectives

## SECTIONS, IN ORDER

- 1 **Heat transfer** — conduction, convection, radiation, and which L8 solves
- 2 **Conduction** — the steady equation, variable  $k$ , the thermal walls, scaling
- 3 **Geometry** — sampling a domain with a hole, its multiplier, its normal
- 4 **Plate with a hole** — verify, solve, weight, and balance the heat

## READ, AND WHAT EACH ANSWERS

|                        |                                      |
|------------------------|--------------------------------------|
| <b>Incropera ch. 1</b> | how does heat move?                  |
| <b>Liu ch. 2.4</b>     | what does a thermal problem need?    |
| <b>Liu ch. 3</b>       | how is a domain with a hole handled? |
| <b>Liu ch. 4.6</b>     | how is the answer checked?           |

# Recap

## 1 – 2 · STEADY

### 1 What decides the type of a second-order equation, and which conditions does each type need?

How it treats time. An elliptic equation has none: a condition on every wall. A parabolic one has one time derivative: the walls and one initial field. A hyperbolic one has two: the walls and two initial conditions. With fewer, the problem has many answers. *Liu ch. 2; Ex\_07.2 notebooks 1 – 3*

### 2 What do Poisson's and Helmholtz's equations describe, and what must a Helmholtz network be protected from?

Poisson is a steady balance between a source and diffusion. Helmholtz adds a term in the field itself, so it has modes, and  $u \equiv 0$  always satisfies it: the network needs a unit-norm term and a floor on  $\lambda$ . In the slot,  $\lambda$  reaching the first eigenvalue is thermal runaway. *Liu ch. 2.4, 6.2; Ex\_07.2 notebook 2*

## 3 – 4 · IN TIME

### 3 What does time add: how do the heat and the wave equation treat a disturbance, and an error made early?

Heat spreads and forgets: every feature decays, the sharpest fastest, and an early error is damped. A wave travels at a finite speed and keeps its shape, so an early error is carried along. Heat needs one initial condition, the wave two. *Liu ch. 4.1, 5.1; Ex\_07.2 notebooks 1, 3*

### 4 How does a PINN meet the conditions of each type, and which type is hardest for it?

A condition with a formula is built into the trial function: the walls, the initial field. One without, like the panel's initial velocity, is a term in the loss. The wave is hardest: its error is not damped, and its residual must be scaled before the network moves. *Liu ch. 4.2, 5.2; Ex\_07.2 notebook 3*

# Conduction

**Conduction** — heat carried through a material by its atoms and electrons, with no motion of the material itself. It flows from hot to cold, at a rate proportional to the temperature gradient: Fourier's law, with the conductivity  $k$  as the property of the material.

## FOURIER'S LAW

$$\mathbf{q} = -k \nabla T$$

$\mathbf{q}$  the heat flux in  $\text{W}/\text{m}^2$ ,  $k$  the conductivity in  $\text{W}/(\text{m K})$ ,  $\nabla T$  the gradient in  $\text{K}/\text{m}$

- **driven by** — a temperature gradient, inside a solid or a still fluid
- **direction** — from hot to cold, hence the minus sign
- **examples** — a chip into its heat sink, a winding into its slot

### CONDUCTION IN PYTORCH

```
q = -k * grad(T, xy)          # Fourier's law: flux from the gradient
```

## CONDUCTIVITY OF COMMON MATERIALS

| material        | $k$ , $\text{W}/(\text{m K})$ |
|-----------------|-------------------------------|
| copper          | 400                           |
| aluminium       | 237                           |
| silicon         | 150                           |
| stainless steel | 15                            |
| glass           | 1                             |
| water           | 0.61                          |
| air             | 0.03                          |

*Typical values at room temperature. Four orders of magnitude lie between copper and air: a heat sink is metal, and insulation is trapped air.*

# Convection

**Convection** — heat carried away from a surface by a moving fluid. The motion of the fluid is not solved for: its effect is put into one coefficient  $h$ , and the flux is proportional to the difference between the surface and the fluid. That is Newton's law of cooling.

## NEWTON'S LAW OF COOLING

$$q = h (T_s - T_\infty)$$

$h$  the heat transfer coefficient in  $\text{W}/(\text{m}^2 \text{ K})$ ,  $T_s$  the surface,  $T_\infty$  the fluid far from it

- **natural** — the fluid moves by itself: warm fluid rises
- **forced** — a fan or a pump moves it, and  $h$  is larger
- **examples** — a heat sink in air, a cold plate with water

### CONVECTION IN PYTORCH

```
q = h * (T_s - T_inf)          # Newton's law: flux from the surface
```

## HEAT TRANSFER COEFFICIENT

| flow           | $h$ , $\text{W}/(\text{m}^2 \text{ K})$ |
|----------------|-----------------------------------------|
| natural, air   | 2 – 25                                  |
| forced, air    | 25 – 250                                |
| natural, water | 50 – 1 000                              |
| forced, water  | 100 – 20 000                            |
| boiling water  | 2 500 – 100 000                         |

*Typical ranges.  $h$  is not a property of the fluid alone: it depends on the speed and on the shape, and comes from a correlation or a measurement.*

# Radiation

**Radiation** — heat exchanged as electromagnetic waves between surfaces that see each other; it needs no material in between. The power a surface radiates grows with the fourth power of its absolute temperature: the Stefan–Boltzmann law, with the emissivity  $\varepsilon$  of the surface.

## STEFAN–BOLTZMANN LAW

$$q = \varepsilon \sigma (T_s^4 - T_{\text{sur}}^4)$$

$\varepsilon$  the emissivity, 0 to 1,  $\sigma = 5.67 \times 10^{-8} \text{ W/(m}^2 \text{ K}^4)$ ,  $T_{\text{sur}}$  the surroundings

- **needs** — no medium: it crosses a vacuum
- **nonlinear** — the fourth power, with  $T$  in kelvin
- **examples** — the sun, a furnace wall, a radiator in space

### RADIATION IN PYTORCH

```
q = eps * sigma * (T_s**4 - T_sur**4)  # Stefan-Boltzmann; T in kelvin
```

## EMISSIVITY OF COMMON SURFACES

| surface            | $\varepsilon$ |
|--------------------|---------------|
| polished aluminium | 0.05          |
| anodised aluminium | 0.80          |
| black paint        | 0.95          |
| a black body       | 1             |

*Typical values. A surface at 80 °C with  $\varepsilon = 0.90$ , in surroundings at 25 °C, radiates 390.50 W/m<sup>2</sup>. Natural convection at  $h = 10 \text{ W/(m}^2 \text{ K)}$  carries 550 W/m<sup>2</sup> from the same surface: the two are of one size.*

# Heat Transfer in L8

**Heat transfer in L8** — of the three mechanisms only conduction is an equation for a field inside the solid, and that field is what a PINN solves for. Convection and radiation act at the surface: they enter as boundary conditions, and their own physics is left out.

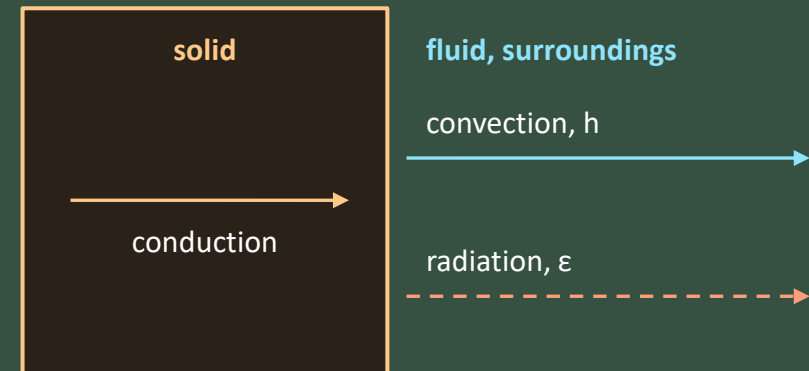
## WHERE EACH MECHANISM ENTERS

|            | acts                       | in L8                        |
|------------|----------------------------|------------------------------|
| conduction | inside the solid           | the equation the PINN solves |
| convection | at a surface, into a fluid | a Robin wall, with $h$ given |
| radiation  | between surfaces           | left out, or counted in $h$  |

## WHY THE OTHER TWO ARE NOT SOLVED

- **convection** — its field is the flow of the fluid: that is L9
- **radiation** —  $T^4$  and the view between surfaces: nonlinear
- **both** — act at the wall, so a boundary condition carries them

## ONE BODY, THREE MECHANISMS



*Radiation is not small: at 80 °C against 25 °C, with  $\varepsilon = 0.90$ , it acts like a coefficient of 7.10 W/(m<sup>2</sup> K), the size of  $h$  in still air. Where it matters, L8 counts it in  $h$ ; it does not solve it.*

# Steady Conduction

**Steady conduction** — Fourier's law in an energy balance: in steady state what leaves a small volume equals what is generated in it,  $\nabla \cdot \mathbf{q} = Q$ . With  $\mathbf{q} = -k \nabla T$  that is Poisson's equation for the temperature, with  $Q/k$  as its source.

## FOURIER'S LAW IN AN ENERGY BALANCE

$$\mathbf{q} = -k \nabla T, \quad \nabla \cdot \mathbf{q} = Q$$

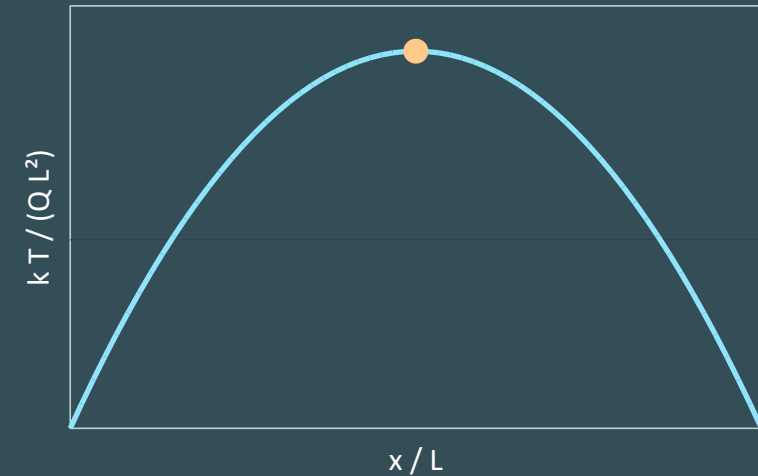
$$T_{xx} + T_{yy} + \frac{Q}{k} = 0$$

$\mathbf{q}$  the heat flux,  $k$  the conductivity in W/(m K)  
 $Q$  the heat generated in W/m<sup>3</sup>

### CONDUCTION IN PYTORCH

```
T = net(xy)                # the temperature, in kelvin
q = -k * grad(T, xy)        # Fourier's law; grad = autograd
F = div(q, xy) - Q          # the energy balance, zero inside
```

## A WALL THAT GENERATES HEAT



*Both faces held at  $T_0$ . The profile is a parabola, and at the amber peak the gradient, and so the flux, is zero: heat leaves through both faces.*

# Variable Conductivity

**Variable conductivity** — when  $k$  changes with position it stays inside the divergence:  $\nabla \cdot (k \nabla T) + Q = 0$ . Across an interface between two materials the gradient jumps and the normal flux does not, so the divergence must not be expanded; autograd differentiates  $k \nabla T$  as written.

## K INSIDE THE DIVERGENCE

$$\nabla \cdot (k(\mathbf{x}) \nabla T) + Q(\mathbf{x}) = 0$$

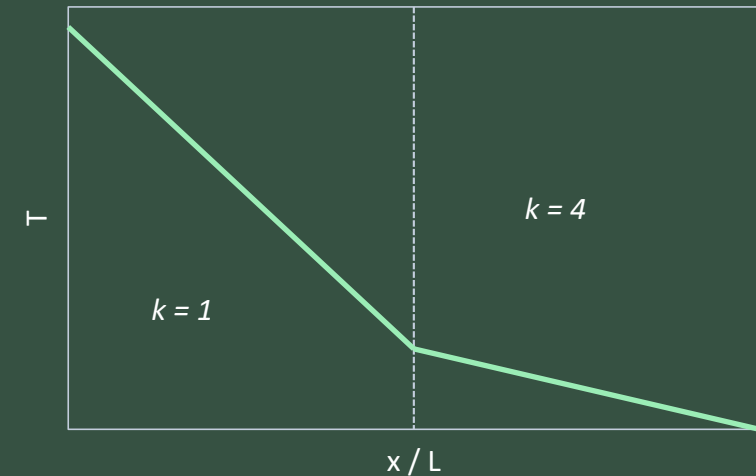
$k(\mathbf{x})$  the conductivity at each point,  $k \nabla T$  differentiated as one quantity

- **at an interface** — the slope jumps, the flux does not
- **in code** — differentiate the product, not its parts

### VARIABLE k IN PYTORCH

```
k = k_field(xy)           # the conductivity at each point
q = -k * grad(T, xy)      # Fourier's flux, one quantity
F = div(q, xy) - Q        # differentiated as written
```

## TWO MATERIALS, ONE INTERFACE



*Left face at 1, right face at 0, no generation. The slope falls fourfold at the interface; the flux  $k dT/dx$  is the same on both sides.*

# Thermal Boundary Conditions

**Thermal boundary conditions** — three ways to fix a wall: a temperature (Dirichlet), a heat flux (Neumann), or convection to a fluid (Robin), which ties the flux to the wall temperature. Only a temperature can be built into the network; the other two stay loss terms.

## A TEMPERATURE, A FLUX, A CONVECTION

### Dirichlet

$$T = \bar{T} \text{ on } \partial\Omega_D$$

### Neumann

$$-k \frac{\partial T}{\partial n} = \bar{q} \text{ on } \partial\Omega_N$$

### Robin

$$-k \frac{\partial T}{\partial n} = h (T - T_\infty)$$

$\bar{T}$  the wall temperature,  $\bar{q}$  the wall heat flux,  $n$  the outward normal  
 $h$  the heat transfer coefficient,  $T_\infty$  the fluid temperature

### THE WALLS IN PYTORCH

```
B_D = T(x_D) - T_bar           # Dirichlet: the temperature
B_N = -k * dTdn(x_N) - q_bar    # Neumann: Fourier's flux, given
B_R = -k * dTdn(x_R) - h * (T(x_R) - T_inf) # Robin: Fourier's flux, tied to T
```

## ROBIN · CONVECTION AT THE WALL

- **balance** — heat conducted to the wall equals heat the fluid carries away
- $h$  — heat transfer coefficient,  $\text{W}/(\text{m}^2 \text{K})$ , or film coefficient: the fluid layer at the wall in one number
- **radiation** — where it matters, it is counted in  $h$

## BIOT NUMBER

$$\text{Bi} = \frac{hL}{k} = \frac{L/k}{1/h}$$

$L$  the size of the body,  $L/k$  resists conduction inside  
 $1/h$  resists convection at the wall

$\text{Bi} \ll 1$ : nearly insulated (Neumann).  $\text{Bi} \gg 1$ : the wall is at  $T_\infty$  (Dirichlet). 1 mm of silicon in still air:  $1.67 \times 10^{-4}$ .

# Well-Posedness

**Compatibility condition** — a problem with a flux condition on every wall has a solution only if the heat out through the walls equals the heat generated, and then only up to a constant: any shift of  $T$  fits the same equation and walls. A PINN picks one without warning.

## HEAT OUT EQUALS HEAT GENERATED

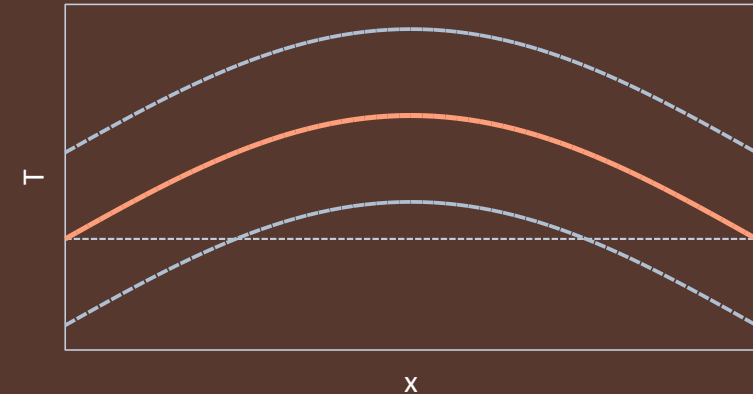
$$\oint_{\partial\Omega} \bar{q} ds = \int_{\Omega} Q d\Omega$$

$\bar{q}$  the flux out through the boundary  $\partial\Omega$ ,  $Q$  the heat generated inside

- **one value pinned** — fixes the constant
- **one wall at a temperature** — fixes it too

**TALK · 2 MIN** How would you explain why a problem with only flux conditions has no single solution, in your own words? Then, in the plot on the right, which of the three curves is right, and what would you add to the loss to choose it?

## THE SAME FIELD, SHIFTED



## FIXING THE LEVEL

```
L_pin = (T(x_ref) - T_ref).pow(2)    # pin one value
L_mean = T(xy).mean().pow(2)         # or fix the mean
```

# Non-Dimensional Form

**Non-dimensional form** — the problem rewritten in scaled variables: lengths divided by the device size  $L$ , temperature by a rise  $\Delta T = QL^2/k$ . Every term is then of order one, as the initialisation and the loss weights assume. Train scaled, and convert back at the end.

## SCALE, TRAIN, CONVERT BACK

$$\theta = \frac{T - T_0}{\Delta T}, \quad \xi = \frac{x}{L}, \quad \Delta T = \frac{QL^2}{k}$$

$\theta$  the scaled temperature,  $\xi$  the scaled coordinate,  $\Delta T$  the temperature scale

$$\theta_{\xi\xi} + \theta_{\eta\eta} + 1 = 0$$

$\eta = y/L$ ; the same equation, every coefficient equal to one

### SCALING IN PYTORCH

```
dT = Q * L**2 / k           # the temperature scale, kelvin
theta = net(xy_m / L)       # train on scaled coordinates
T = T0 + dT * theta         # convert back, in kelvin
```

## A SILICON DEVICE, BEFORE AND AFTER

|                  | physical                | scaled                     |
|------------------|-------------------------|----------------------------|
| length $L$       | 1 mm                    | 1                          |
| conductivity $k$ | 150 W/(m K)             | 1                          |
| source $Q$       | $10^8$ W/m <sup>3</sup> | 1                          |
| rise $\Delta T$  | $QL^2/k = 0.67$ K       | 1                          |
| coefficient $h$  | 25 W/(m <sup>2</sup> K) | $Bi = 1.67 \times 10^{-4}$ |

*In metres and kelvin the source term is ten to the eighth and the length ten to the minus third. Scaled, every term is of order one.*

# Sampling a Real Domain

**Rejection sampling** — points drawn uniformly over a box around the domain, keeping those the level set places inside,  $\phi > 0$ . On a curved wall the points are spaced by arc length rather than by angle, so the ends of the hole, where it bends most, get their share.

## ONE FORMULA: INSIDE, OUTSIDE, ON THE WALL

$$\phi(x, y) = \left(\frac{x - x_c}{a}\right)^2 + \left(\frac{y - y_c}{b}\right)^2 - 1$$

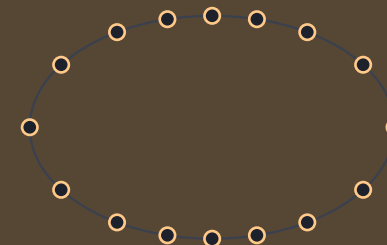
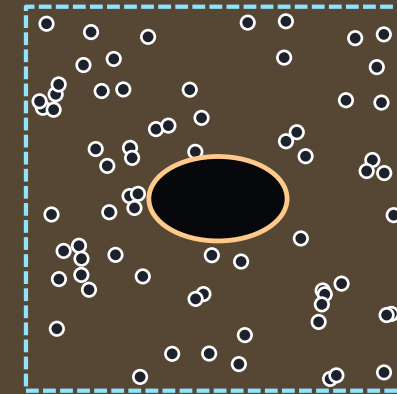
$(x_c, y_c)$  the centre,  $a, b$  the half axes;  $\phi < 0$  in the hole,  $\phi = 0$  on its wall

- **inside** — keep a point where  $\phi > 0$
- **on the wall** — equal steps of arc length

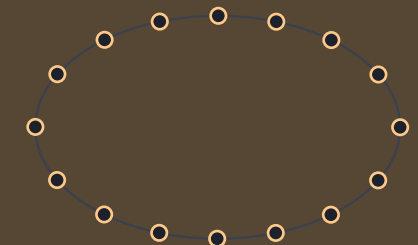
### SAMPLING IN PYTORCH

```
xy = torch.rand(4000, 2)           # uniform over the square
xy = xy[phi(xy) > 0][:2000]        # keep the points in the plate
xy_h = ellipse_by_arc_length(160)  # the hole wall, evenly spaced
```

## THE PLATE, AND ITS WALL POINTS



equal angle: ends bare



equal arc length

# Boundary Multiplier

**Boundary multiplier** — a function that is zero on a boundary and positive in the domain. Multiplied into the network's output, it makes a value hold there exactly. For an ellipse the level set  $\phi$  is one; for a shape with no formula, a small second network is trained to be one, then frozen.

## THE HOLE WALL, BUILT IN

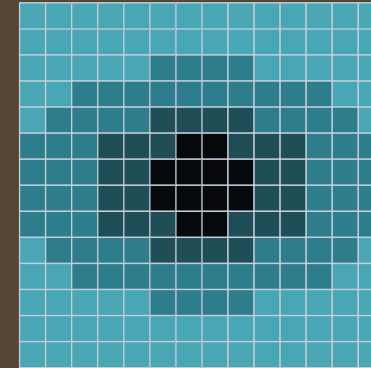
$$T(\mathbf{x}) = \phi(\mathbf{x}) \mathcal{N}_{\hat{w}}(\mathbf{x})$$

$\phi$  the level set, zero on the hole;  $\mathcal{N}_{\hat{w}}$  the network;  $T = 0$  there for any weights

- **a value** — held exactly by a factor
- **a derivative** — cannot be, stays a loss term

**TALK · 2 MIN** How would you explain what a boundary multiplier does, in your own words? Then, in the figure on the right, where is the multiplier small, and what does that do to the network's gradient near the hole?

## $\phi$ OVER THE PLATE



dark: the hole

## THE HOLE IN PYTORCH

```
T = hole_multiplier(xy) * net(xy)    # zero on the hole
loss = F.pow(2).mean() + w_N * L_N  # edges still a term
```

# Flux on a Curved Boundary

**Normal derivative** — the rate of change of  $T$  across a wall,  $\partial T / \partial n = \nabla T \cdot \mathbf{n}$ , where  $\mathbf{n}$  is the unit normal pointing out of the material. On a curved wall  $\mathbf{n}$  changes with position: it comes from the gradient of the level set, and its sign decides whether heat leaves or enters.

## THE NORMAL FROM THE LEVEL SET

$$\frac{\partial T}{\partial n} = \nabla T \cdot \mathbf{n}, \quad \mathbf{n} = -\frac{\nabla \phi}{|\nabla \phi|}$$

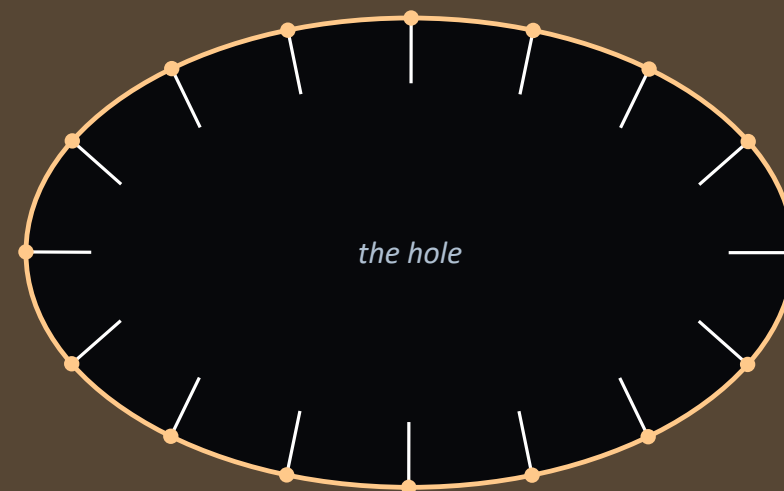
$\mathbf{n}$  the unit normal out of the plate; minus, because out of the plate is into the hole

- **on an edge** — the normal is a constant
- **on the hole** — it turns with the wall

## THE NORMAL IN PYTORCH

```
g = grad(phi(xy_h), xy_h)           # the level set's gradient
n = -g / g.norm(dim=1, keepdim=True) # unit normal, out of the plate
q_n = -k * (grad(T, xy_h) * n).sum(1) # Fourier's flux through the wall
```

## NORMALS ON THE HOLE WALL



*Each white stick is  $\mathbf{n}$  at an amber point: it turns with the wall and always points out of the plate. A sign error inverts the flux, not the magnitude.*

# Manufactured Solution

**Manufactured solution** — a solution chosen in advance, with the source derived from it by differentiation, so the answer is known exactly. Solving that problem checks the residual, the trial solution and the training loop before they meet a problem with no answer to compare against.

## CHOOSE THE ANSWER, DERIVE THE SOURCE

$$T = \sin(\pi x)\sin(\pi y) \Rightarrow \frac{Q}{k} = 2\pi^2 \sin(\pi x)\sin(\pi y)$$

$T$  the chosen answer, zero on the four edges;  $Q/k$  the source derived from it

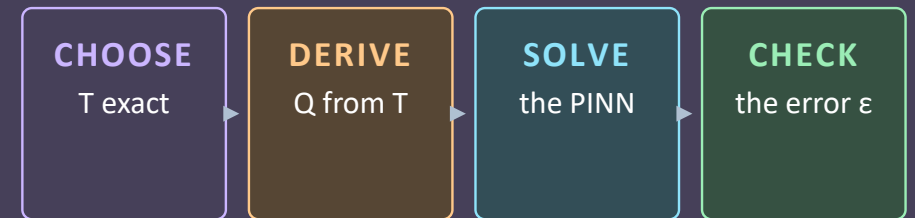
$$\varepsilon = \frac{\|T_{\hat{w}} - T_{\text{exact}}\|_2}{\|T_{\text{exact}}\|_2}$$

$T_{\hat{w}}$  the network's answer,  $T_{\text{exact}}$  the chosen one

### THE CHECK IN PYTORCH

```
Q_k = 2 * pi**2 * sin(pi*x) * sin(pi*y)    # the source, from the answer
T = x*(1-x) * y*(1-y) * net(xy)           # the walls held exactly
err = (T - T_exact).norm() / T_exact.norm() # compare with the answer
```

## FOUR STEPS BEFORE THE PLATE



**what it checks**

the residual, the trial, the loop

**what it cannot**

the geometry of the plate

*An error found here costs minutes; the same error found on the plate costs an afternoon.*

# Heated Plate with a Hole

**Heated plate with a hole** — the case study of this block: a square plate with uniform heat generation, cooled through an elliptical hole held at  $T = 0$ , with insulated outer edges,  $\partial T / \partial n = 0$ . The hole wall is built into the network by its level set; the edges stay a loss term.

## ONE BUILT-IN WALL, ONE LOSS TERM

$$\mathcal{L} = \mathcal{L}_{PDE} + w_N \mathcal{L}_{NBC}$$

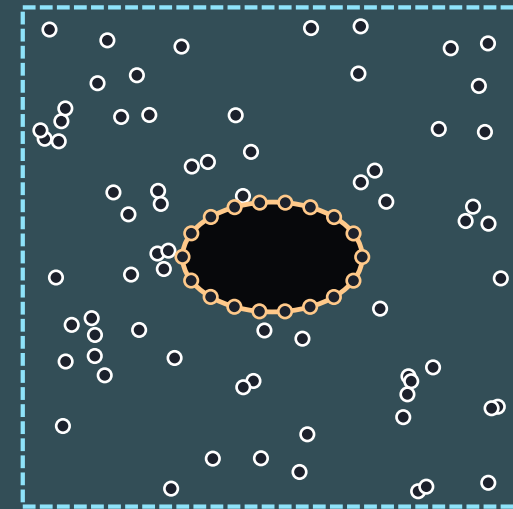
$\mathcal{L}_{PDE}$  the residual inside,  $\mathcal{L}_{NBC}$  the insulated edges,  $w_N$  their weight

- **the hole** —  $T = 0$ , by the multiplier
- **the edges** — no flux, a weighted term

### THE PLATE IN PYTORCH

```
T = hole_multiplier(xy) * net(xy)          # the hole held at zero
L_N = dTdn(T, xy_edges).pow(2).mean()      # the edges: no flux
loss = (lap(T) + Q_k).pow(2).mean() + w_N * L_N # residual and edges
```

### THE PLATE OF EX\_08.1



**amber wall**

$T = 0$ , built in

**dashed edges**

$\partial T / \partial n = 0$ , a loss term

# Flux Weight

**Flux weight** — the weight  $w_N$  on the insulated-edge term. Too small, and the network ignores the edges and heat leaks through them; too large, and the residual inside pays for a perfect wall. It is chosen by a sweep that watches both terms, not the total loss.

## ONE RUN PER WEIGHT, BOTH TERMS WATCHED

$$\mathcal{L} = \mathcal{L}_{PDE} + w_N \mathcal{L}_{NBC}$$

$\mathcal{L}_{PDE}$  the residual inside,  $\mathcal{L}_{NBC}$  the insulated edges,  $w_N$  their weight

- **the total loss** — hides which term is paying
- **the choice** — a trade, argued from both

### THE SWEEP IN PYTORCH

```
for w_N in [0.1, 1, 10, 100]:
    net = train(w_N)
    log(w_N, L_pde(net), L_N(net))
```

# one run per weight  
# the same seed, the same points  
# both terms, not their sum

## WHAT EACH END OF THE SWEEP DOES

**w\_N small**

the edges leak heat

**w\_N large**

the residual inside pays

**between**

argued from both terms

*The weight is a rate of exchange between a flux and a residual, which have different units. Scaling changes the rate before any sweep does.*



# Energy Balance Check

**Energy balance check** — a check on the trained network, not a new law. Fourier's law gives the heat flux at each point of the hole's wall; added up around the hole, that is the heat leaving the plate. In steady state it equals the heat generated: the energy balance, for the whole plate.

## FOURIER'S FLUX, ADDED UP AROUND THE HOLE

$$P_h \overline{q_n} = Q A_\Omega$$

$P_h$  the hole's perimeter,  $\overline{q_n}$  the mean of Fourier's flux into it,  $A_\Omega$  the plate's area

- **at a point** — Fourier's law gives the flux into the hole
- **around the hole** — its mean times the perimeter is the heat out
- **in the plate** —  $Q$  times the area is the heat generated

### THE BALANCE IN PYTORCH

```
q_n = -k * (grad(T, xy_h) * n).sum(1)    # 1: Fourier's flux at each wall point
out = q_n.mean() * perimeter              # 2: heat out; points evenly spaced
gen = Q * (1 - pi * a * b)                # 3: heat generated; compare with out
```

## THE CHECK IN NOTEBOOK 03

|                      | value   |
|----------------------|---------|
| heat generated       | 0.94    |
| heat out             | 0.92    |
| heat out / generated | 98.57 % |
| out short by         | 1.43 %  |

*The plate with a hole has no exact solution to compare against, so this balance is the check: heat generated is known, heat out is the network's. Notebook 03 calls it the flux balance. Scaled units; the ratio is taken before rounding. Whether 1.43 % is small enough is your decision.*

# Advanced Topics

RELATED TOPICS LEFT OUT, AND WHERE THEY MATTER

| topic                      | what it adds                          | read            | where it makes a difference |
|----------------------------|---------------------------------------|-----------------|-----------------------------|
| energy method              | an energy, first derivatives only     | Liu ch. 2.8     | elasticity, structures      |
| triangulated domain        | points and weights from a mesh        | Liu ch. 3.2     | any drawn geometry          |
| convergence study          | error against points and size         | Liu ch. 4.9.3   | a design sign-off           |
| <u>diffusion surrogate</u> | trained on solved cases, then steered | UDL ch. 14 – 18 | many boundary conditions    |
| PINN or FEM                | when each is the better tool          | Liu ch. 2.10    | one-off steady solves       |

WHY THEY ARE LEFT OUT

The plate of Ex\_08.1 needs none of them: its hole has a formula, its answer can be checked by an energy balance, and it is solved once. A finite-element solver would solve it too; the method earns its place on the inverse problems of L8.2.

# Exercise 8.1

## THE NOTEBOOKS, IN ORDER

- **00** — the plate, its hole, its samples, its normals
- **01** — the method, on a manufactured solution
- **02** — the plate with its hole built in
- **03** — the flux weight, and the energy balance
- **04** — compares them, and the report

## THE QUESTION EACH NOTEBOOK ANSWERS

- 01** is the method right?
- 02** how hot, and where?
- 03** does the heat balance?

## REPORT QUESTION, TO CONCLUDE

The manufactured solution checked the method and the energy balance checked the plate. On a design with no exact solution, which checks would you run, and what result would make you distrust the network?

# Key Takeaways

## HEAT TRANSFER

|                    |                                     |
|--------------------|-------------------------------------|
| <b>what it is</b>  | conduction, convection, radiation   |
| <b>function</b>    | T by conduction; the rest are walls |
| <b>train, tune</b> | scale first; flux walls weighted    |
| <b>vs Poisson</b>  | units, k inside, three walls        |
| <b>used for</b>    | devices, plates, heat sinks         |

## REAL GEOMETRY

|                     |                                     |
|---------------------|-------------------------------------|
| <b>what it is</b>   | a domain with no polynomial factor  |
| <b>function</b>     | $\phi$ : inside, multiplier, normal |
| <b>train, tune</b>  | arc length on the wall; $w_N$       |
| <b>vs rectangle</b> | n and the factor vary with x        |
| <b>used for</b>     | plates with holes, device sections  |

*Next — L8.2: the same plate, in time.*

# Questions

## 1 – 2 · HEAT TRANSFER

### 1 What are the three mechanisms of heat transfer, and which of them does a PINN solve in L8?

Conduction through a material (Fourier's law), convection from a surface into a moving fluid (Newton's law of cooling, with  $h$ ), and radiation between surfaces (Stefan–Boltzmann,  $T^4$ ). Only conduction is a field equation inside the solid; the other two act at the surface and enter as boundary conditions. *Incropera ch. 1*

### 2 What does a steady conduction problem need before a PINN solves it, and how is the result checked?

Fourier's law in an energy balance gives Poisson's equation, with  $k$  inside the divergence. Each wall carries a temperature, a flux or convection; with a flux on every wall the level is not fixed. Scale it so every term is of order one, then check by a manufactured solution and the energy balance. *Liu ch. 2.2, 2.4, 4.4; Ex\_08.1 notebooks 01 – 03*

## 3 – 4 · REAL GEOMETRY

### 3 How is a domain with a hole sampled, and how is a temperature built into its wall?

Points are drawn over a box and kept where the level set  $\phi$  is positive; on the curved wall they are spaced by arc length. A boundary multiplier, zero on the wall and positive inside, multiplies the network's output, so the wall temperature holds exactly. *Liu ch. 3.1 – 3.3; Ex\_08.1 notebooks 00 – 02*

### 4 How is a flux condition imposed on a curved wall, and how is its weight chosen?

As a loss term on the normal derivative, the gradient of  $T$  along the outward unit normal, which comes from the level set. Its weight is found by a sweep that watches the flux term and the residual separately: too small and heat leaks through the wall, too large and the interior pays. *Liu ch. 2.2.3, 3; Ex\_08.1 notebook 03*

NEXT

## L8.2 — Transient Heat

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The same plate, in time: an initial field, causality, and a diffusivity recovered from sensors.

### BEFORE THE NEXT SESSION

|             |                                     |
|-------------|-------------------------------------|
| read        | Liu ch. 3, 4.6                      |
| run         | Ex_08.1 notebooks 00 to 03          |
| bring       | the energy balance, and your $w_N$  |
| think about | which of your walls can be built in |

*The steady plate is solved and checked. Next it heats up.*